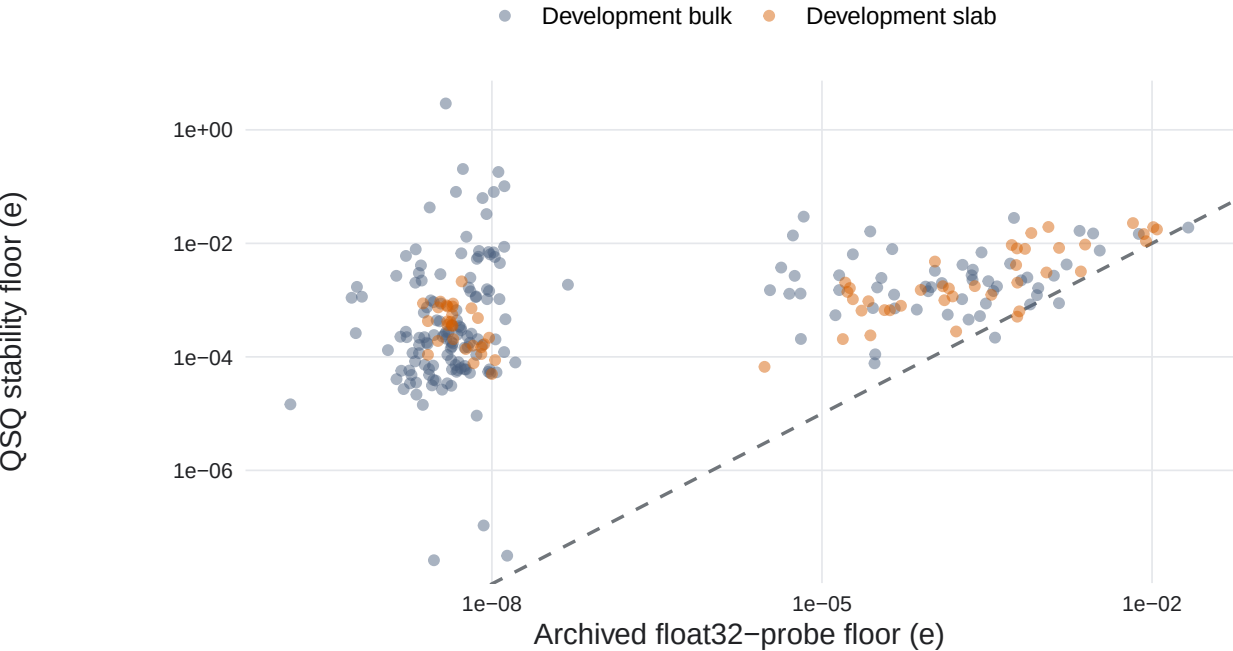


Supplementary Figure S1 | QSQ defines the measurable Bader–fidelity landscape

The order–preserving round–trip control is a method–validation control; current eligibility and certification use QSQ.

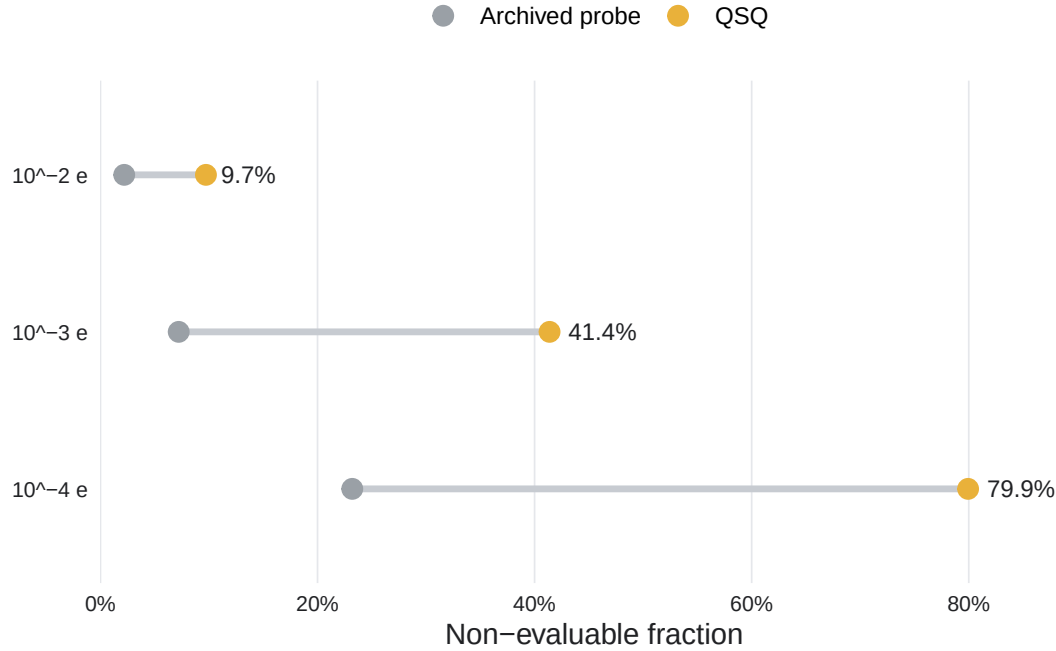
A | QSQ exposes substantially larger Bader instability

Paired order–preserving control vs five–seed QSQ floors; n = 254



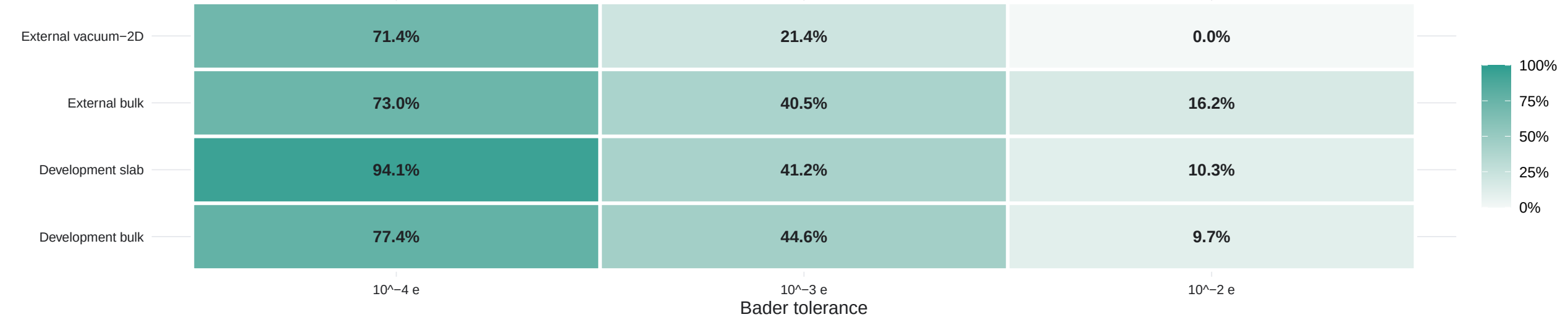
B | Stability qualification changes the non–evaluable rate

Overall 319–system stability universe



C | Numerical eligibility is material– and tolerance–dependent

QSQ; the structural–class contrast is not universal



A, paired order–preserving–control and QSQ floors. B, overall non–evaluable fractions. C, QSQ rates by analysis stratum. Non–evaluable is neither codec pass nor codec failure.